

O'REILLY®

Databricks Fundamentals Bootcamp

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Day 1 of 2



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Learning Objectives

- Set up and navigate the Databricks platform
- Use Apache Spark to build ETL pipelines
- Reliably storing the data using Delta format
- Use Databricks Unity Catalog to manage and govern data
- Build production-ready pipelines using Databricks Workflows



Course Schedule – Day 1

Exploring, Cleaning, and Transforming Data with Databricks

- Introduction to Databricks
- Understanding Apache Spark architecture
- Setting up and exploring Databricks
- Reading data from multiple file formats
- Cleaning and transforming data using PySpark
- Working with Spark SQL and visualizing data



Course Schedule – Day 2

Storing Data with Delta Lake and Building Workflows with Databricks

- Introduction to Delta Lake
- Storing data in data lake using Delta format
- Working with Delta Lake features
- Working with Unity Catalog
- Building workflows with Databricks



Prerequisites

- Azure Subscription for setting up Azure Databricks
OR Databricks Free Edition



Poll: What Best Describes Your Data Profile?

- Data Engineer
- Analytics Engineer / BI Developer
- Data Analyst
- Data Architect
- Data Scientist
- Business User
- Others





Introduction to Databricks

**Extremely fast and powerful
analytics engine for large-scale
data processing.**

Databricks

Unified Data Intelligence Platform

- Build and maintain enterprise-grade **data, analytics, and AI** solutions at scale

SQL

 SQL Editor

 Queries

 Dashboards

 Genie

 Alerts

 Query History

 SQL Warehouses

Data Engineering

 Job Runs

 Data Ingestion

AI/ML

 Playground

 Experiments

 Features

 Models

 Serving

Databricks

Unified Data Intelligence Platform

- Build and maintain enterprise-grade **data, analytics, and AI** solutions at scale
- Managed platform based on Apache Spark - setup infra with few clicks

Compute > New compute > Simple form: ON

Create new compute

General

Compute name
Demo Cluster

Policy
Unrestricted

Performance

☐ Machine learning

Databricks runtime
16.4 LTS (Scala 2.12) Scala 2.12, Spark 3.5.2

☒ Photon acceleration

Worker type
Standard_D4ds_v5 16 GB Memory, 4 Cores

Min Max
1 2

☐ Single node

☒ Enable autoscaling

☒ Terminate after 120 minutes of inactivity

Advanced performance

Create Cancel

Databricks

Unified Data Intelligence Platform

- Build and maintain enterprise-grade **data, analytics, and AI** solutions at scale
- Managed platform based on Apache Spark - setup infra with few clicks
- Integrated workspace to write code & collaborate

The screenshot shows the Databricks SQL interface. At the top, there's a tab labeled 'Medallion Architecture - UC'. Below the tab, there's a toolbar with icons for SQL, Tabs, and other functions. The main area displays a SQL query for a merge operation. The query is as follows:

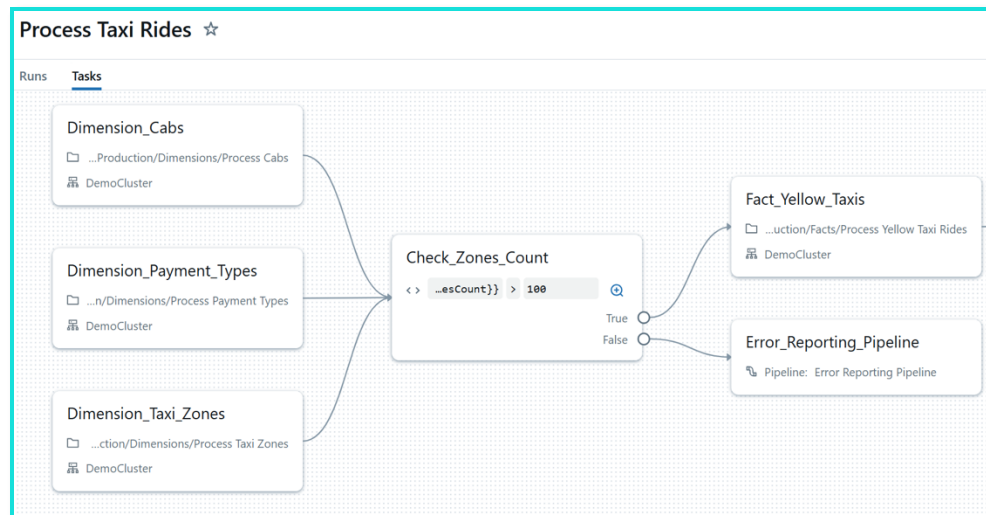
```

MERGE INTO taxidata.dbo.YellowTaxis_Silver tgt
  USING BronzeChanges src
  ON tgt.VendorId = src.VendorId AND tgt.RideId =
    src.RideId
  -- Update row if join conditions match
  WHEN MATCHED
  THEN
    UPDATE SET tgt.PickupTime = src.PickupTime, tgt.DropTime = src.DropTime, tgt.
      PickupLocationId = src.PickupLocationId,
      tgt.DropLocationId = src.DropLocationId, tgt.CabNumber = src.
        CabNumber,
      tgt.DriverLicenseNumber = src.DriverLicenseNumber, tgt.PassengerCount
        = src.PassengerCount,
      tgt.TripDistance = src.TripDistance, tgt.TotalAmount = src.
  
```

Databricks

Unified Data Intelligence Platform

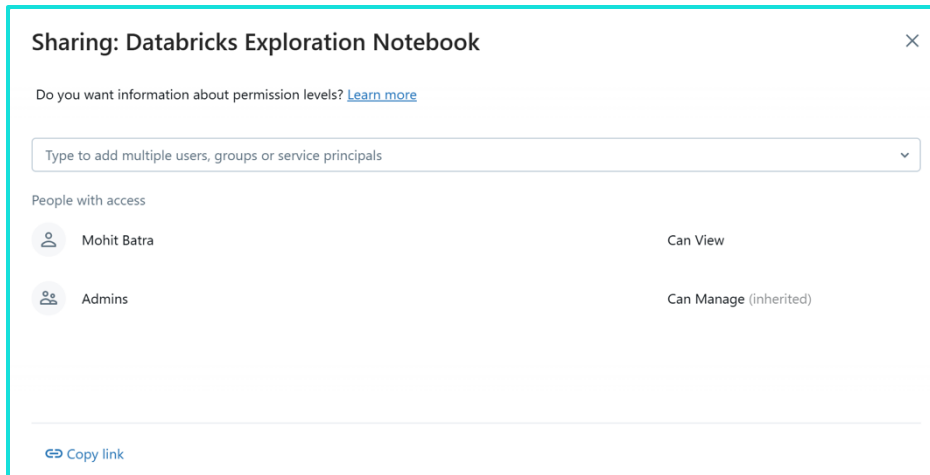
- Build and maintain enterprise-grade **data, analytics, and AI** solutions at scale
- Managed platform based on Apache Spark - setup infra with few clicks
- Integrated workspace to write code & collaborate
- Deploy and schedule jobs



Databricks

Unified Data Intelligence Platform

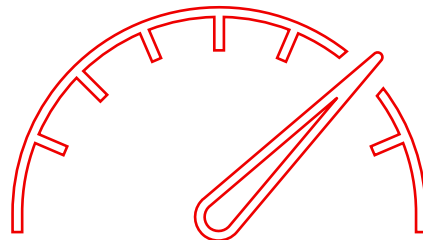
- Build and maintain enterprise-grade **data, analytics, and AI** solutions at scale
- Managed platform based on Apache Spark - setup infra with few clicks
- Integrated workspace to write code & collaborate
- Deploy and schedule jobs
- Enterprise grade security



Databricks

Unified Data Intelligence Platform

- Build and maintain enterprise-grade [data](#), [analytics](#), and [AI](#) solutions at scale
- Managed platform based on Apache Spark - setup infra with few clicks
- Integrated workspace to write code & collaborate
- Deploy and schedule jobs
- Enterprise grade security
- Built-in optimizations – Up to 50x higher performance than vanilla Spark



Databricks



Data Engineering

AI and ML

Data Warehousing

OLTP Databases



Compute Options

Create clusters or use serverless compute

Photon
Engine



Unity Catalog

Unified security, cataloguing and governance



Delta Lake

Storage layer to bring reliability to Data Lakes



AI Integration



Monitoring



Security



Microsoft Azure



Amazon Web Services



Google Cloud



Data Engineering

- **Code using Notebooks**
 - Write Spark code in multiple languages to transform data
 - Build batch or streaming pipelines
- **Lakeflow Connect (Ingestion Pipeline)**
 - Ingest data from variety of data sources
 - Connectors from Databricks and Fivetran are available
- **Lakeflow Declarative Pipelines (ETL Pipeline)**
 - ETL framework to build declarative data pipelines
 - Use it to perform transformations, apply data quality checks, etc.
- **Lakeflow Jobs (Orchestration Jobs)**
 - Orchestrate notebooks, pipelines, queries, and more





AI & ML

- **Machine Learning**

- Train and deploy your own ML models
- Manage complete data and model lifecycle using MLOps

- **AI**

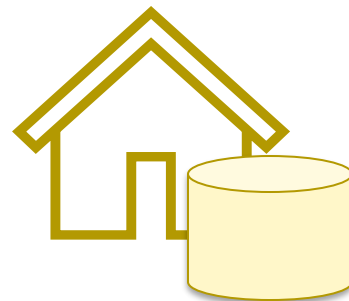
- Use [Databricks Assistant](#) to generate code, fix bugs, etc.
- Supports foundational GenAI LLM models, like OpenAI GPT, Meta Llama, etc.
- Use playground to test models
- Use GenAI models on top of your own data
- Use [Databricks Genie](#) to ask questions on your own data





Data Warehousing

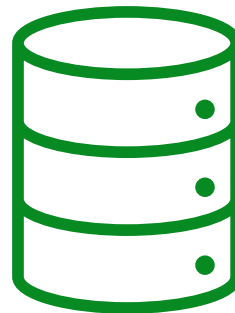
- **Databricks SQL**
 - Collection of services that bring data warehousing capabilities and performance to your existing data lakes
 - Build dashboards and reports





OLTP Databases

- **Lakebase**
 - Fully managed Postgres database for transactional workloads



Question: Which cloud platforms support Databricks?

- Only AWS
- Only Azure
- Only Google Cloud
- AWS and Azure
- AWS, Azure, Google Cloud





Question: Which storage system does Databricks use to store data?

- Its own proprietary storage only
- Cloud object storage like S3, ADLS, GCS
- Local disk only
- Hadoop HDFS exclusively





Understanding Apache Spark Architecture



Apache Spark



- Open-source compute engine to perform distributed data processing at-scale
- Based on cluster computing – Data is processed in the memory of the cluster
- Support for multiple languages – Scala, Python, SQL, R & Java
- Supports multiple use cases – Batch processing, streaming processing, ML, Data Science

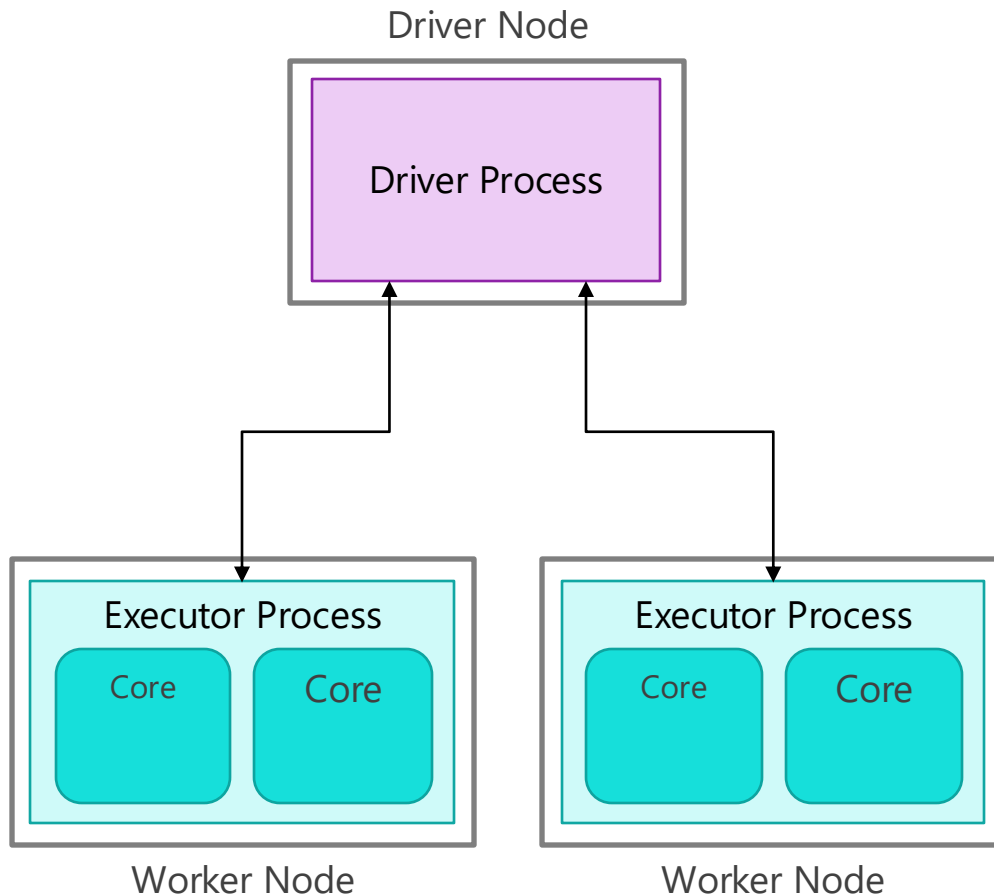


Spark Architecture



1. Read a file from Data Lake
2. Apply transformations
3. Write processed data to Data Lake

- Driver & Executors = JVM processes
- One Worker Node can have one or more Executor processes
- Executor size = 2 cores + 6 GB RAM





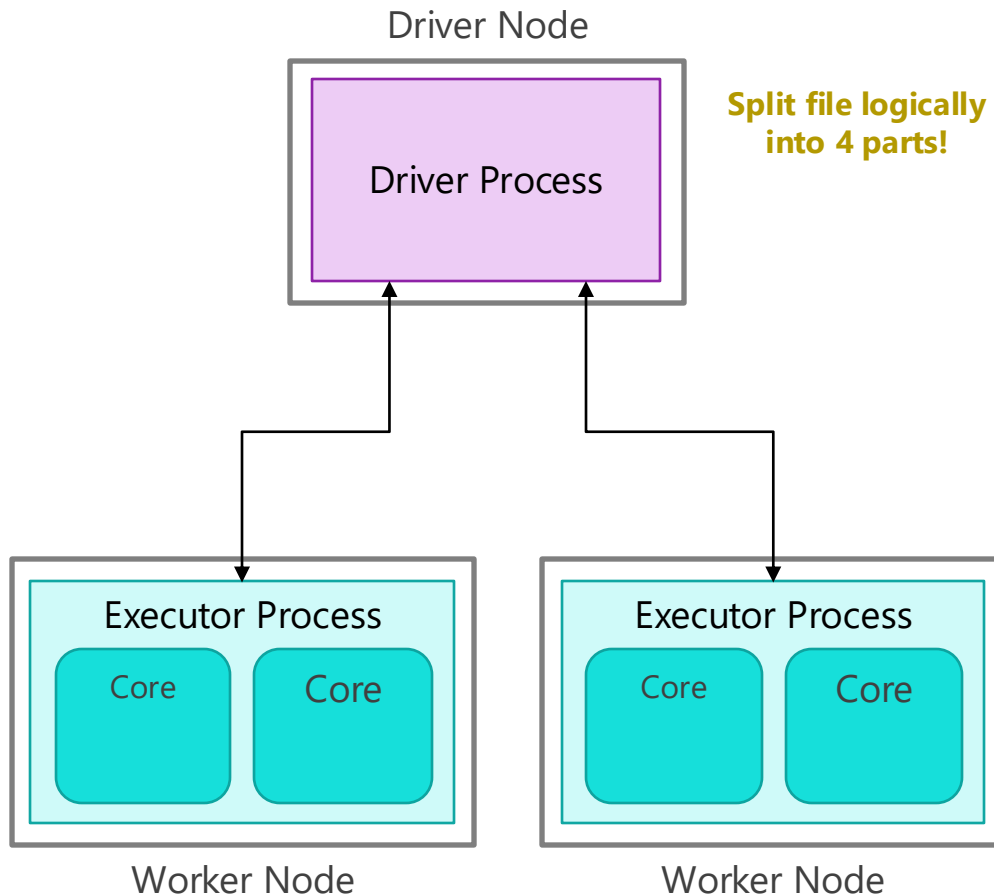
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Data Lake





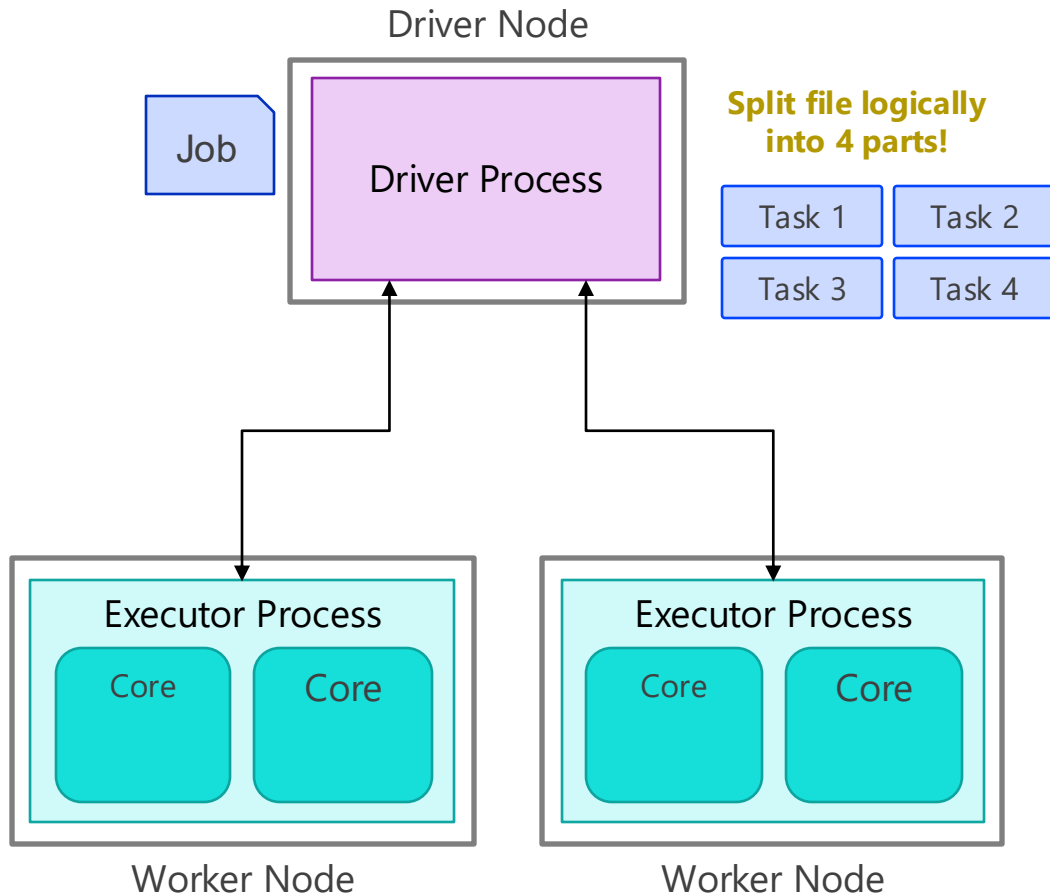
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Data Lake





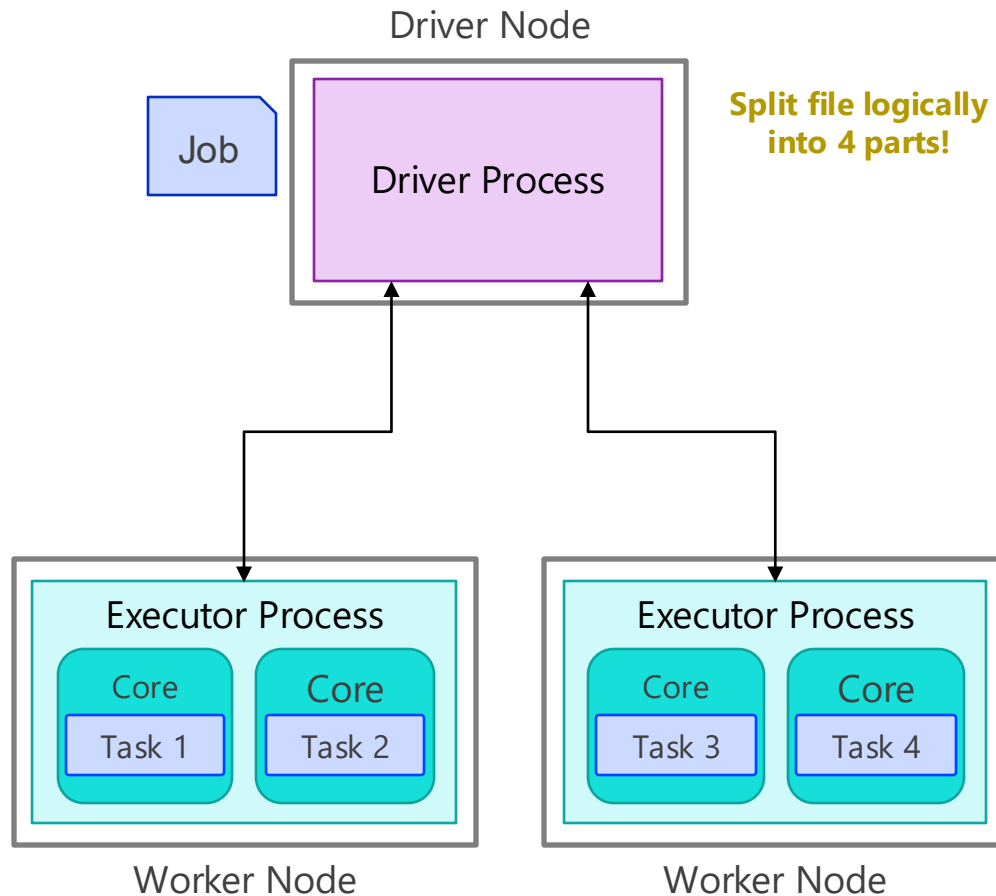
Spark Architecture



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Data Lake

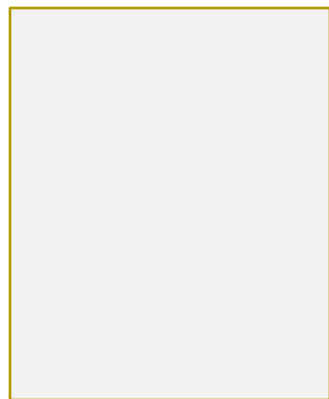




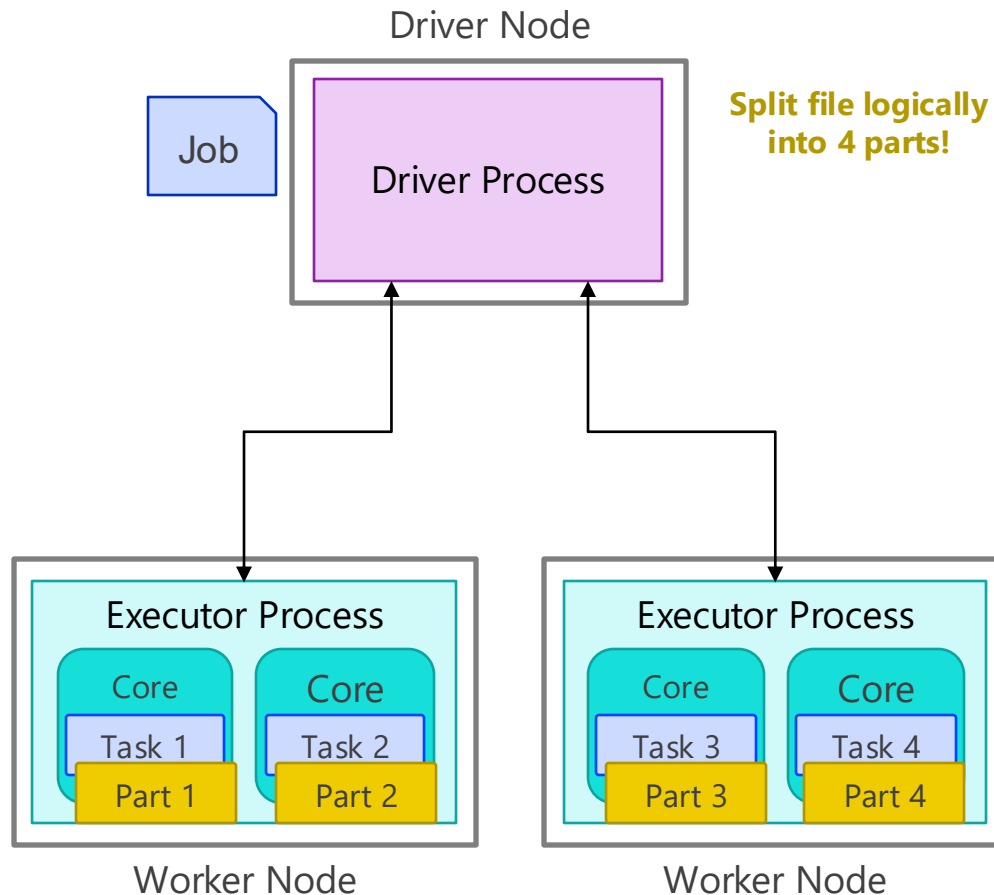
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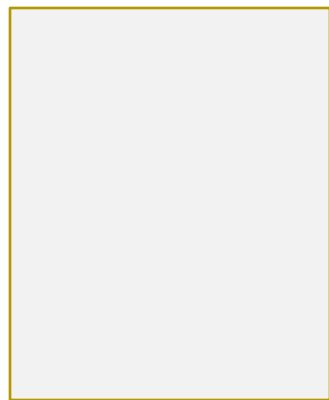




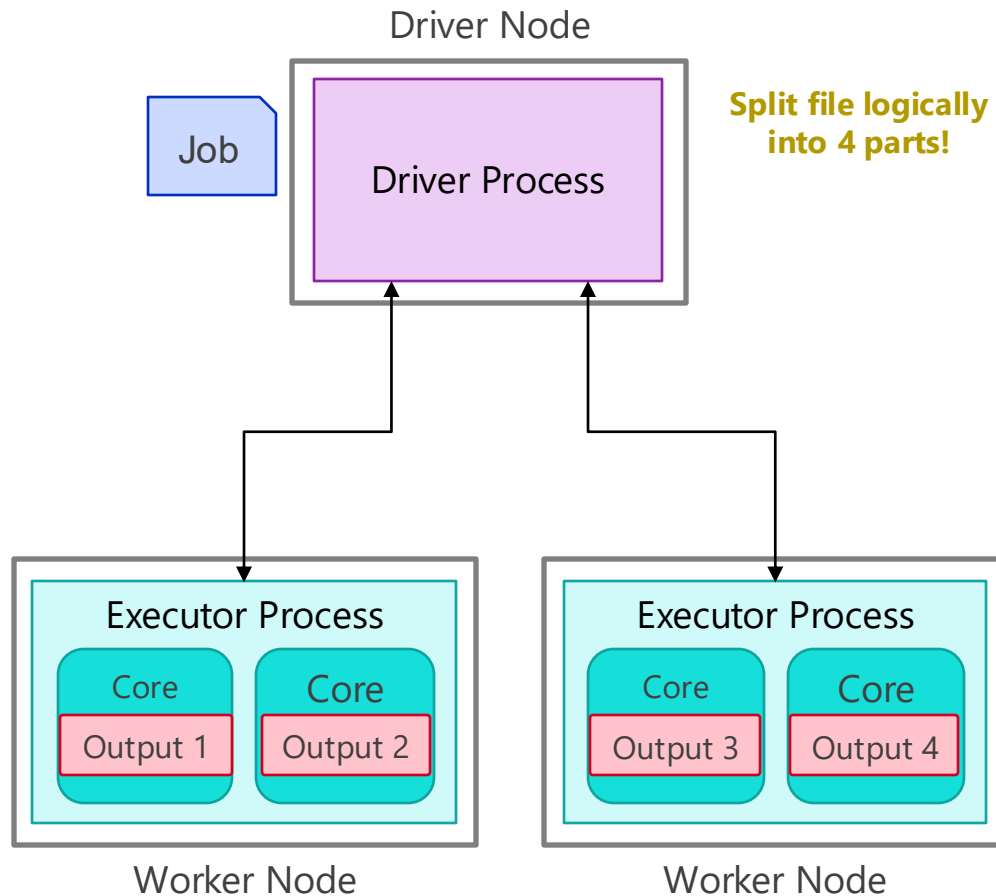
Spark Architecture



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Data Lake

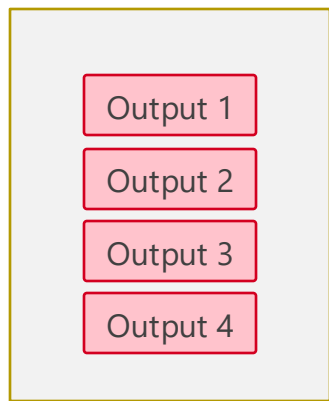




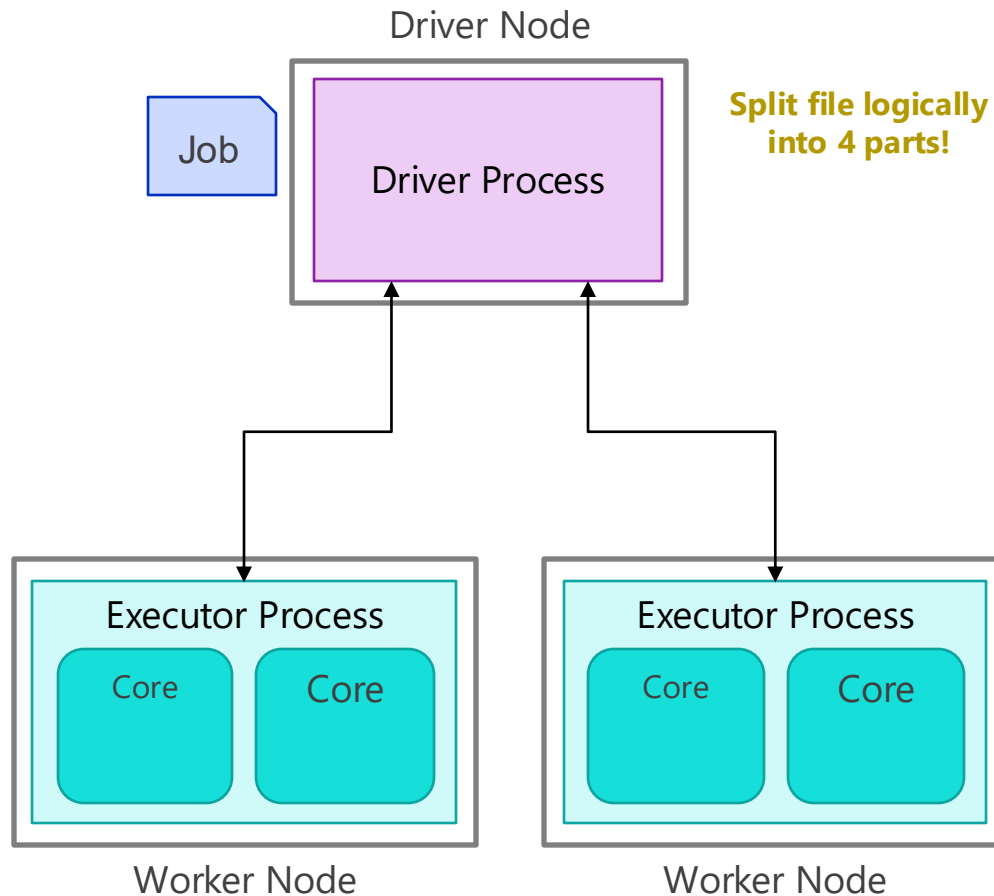
Spark Architecture



1. Read a file from Data Lake
2. Apply transformations
3. Write processed data to Data Lake



Data Lake



Question: Which of the following describes the relationship between nodes and executors?

- Nodes and executors are not related
- Node is a process that runs on an executor
- Executor is a process that runs on a node
- There are always the same number of nodes and executors





Question: In Spark, what type of processes are the driver and the executors?

- Python process
- Java process
- C++ process





Setting Up and Exploring Databricks



Compute Options

- **All-purpose compute**
 - Use it interactively analyze the data using notebooks
 - Keeps running unless manually terminated or auto-terminates on inactivity
- **Jobs compute**
 - Use it to run automated jobs
 - Starts and terminates with the job
- **SQL warehouses**
 - Use it to run SQL commands in Databricks SQL
- **Serverless**
 - On-demand scalable compute
 - Can be used with notebooks, jobs or as a SQL warehouse



Question: In which scenario a data engineer will want to use a Job cluster instead of an all-purpose cluster?

- An ad-hoc analytics report needs to be developed while minimizing compute costs.
- A data team needs to collaborate on the development of a machine learning model.
- An automated workflow needs to be run every 30 minutes.
- A Databricks SQL query needs to be scheduled for upward reporting.





Exercise

- Setup Databricks workspace
 - Create all-purpose cluster (use Serverless with Databricks Free Edition)
 - Upload data files
 - Use *dbutils* to work with file system
-
- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
 - Data Files @ <https://tinyurl.com/oreilly-databricks-bootcamp>

Reading Data from Multiple File Formats



DataFrames

- Higher-level Spark API to write code
- Reads data from source as a tabular structure
- Like a table in relational database – so you can apply typical table type operations

Id	Name	Salary
1	Mohit	20000
2	Ivan	27500
3	Sabrina	24200
4	Andrew	19000
5	Neha	32000

Row

Column



Cleaning and Transforming Data using PySpark



Exercise

- Work with multiple file formats like CSV, JSON and Parquet
- Read files from storage using Spark's DataFrame API
- Apply operations to clean and transform data

- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
- Data Files @ <https://tinyurl.com/oreilly-databricks-bootcamp>



Question: Which command can you use to preview a CSV file's content by reading top few bytes in Databricks?

- `spark.read.csv`
- `sc.getFileBytes`
- `dbutils.fs.head`





Question: Which Spark method can you use on DataFrame to create a new column?

- addColumn
- na.fill
- withColumn





Working with Spark SQL and Visualizing Data



Question: You have created a DataFrame in a notebook using PySpark. In the same notebook, you want SQL to access that DataFrame. What will you do?

- In SQL command, use the DataFrame name as the view name
- Write the DataFrame contents to Data Lake first, and then read it from Data Lake using SQL command
- First create a Temp View, and then use it in SQL command





Exercise

- Run SQL queries on DataFrames
- Create visualizations and build dashboard
- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
- Data Files @ <https://tinyurl.com/oreilly-databricks-bootcamp>



Loading Processed Data to Data Lake



Ways to Store Data

Directly to Data Lake

Files are stored in Data Lake
No metadata is stored

Stored as Spark Table

Files are stored in Data Lake
Metadata is stored in Hive



Exercise

- Write processed data to Data Lake
- Store dataset as Spark tables
- Instructions @ <https://github.com/Crystal-Talks/DatabricksFundamentalsBootcamp>
- Data Files @ <https://tinyurl.com/oreilly-databricks-bootcamp>

Key Takeaways – Day 1

Exploring, Cleaning, and Transforming Data with Databricks

- Introduction to Databricks
- Understood Apache Spark architecture
- Set up and explored Databricks
- Read data from multiple file formats
- Cleaned and transformed data using PySpark
- Worked with Spark SQL and visualized data





Tomorrow...

Storing Data with Delta Lake and Building Workflows with Databricks

- Introduction to Delta Lake
- Storing data in data lake using Delta format
- Working with Delta Lake features
- Working with Unity Catalog
- Building workflows with Databricks



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Feel free to reach out...



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